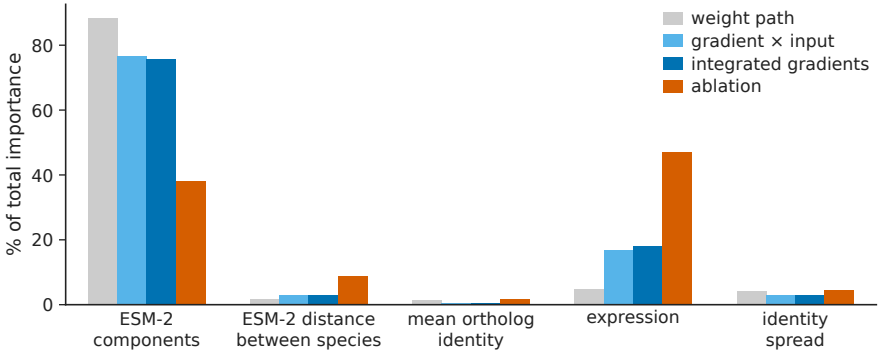
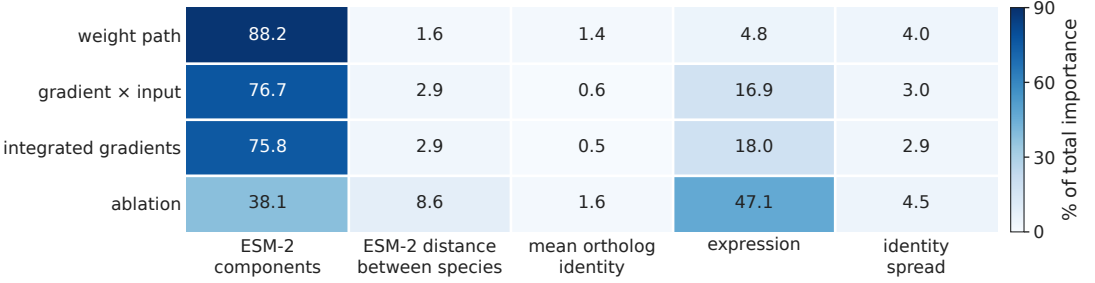

weight path
gradient × input
integrated gradients
ablation
% of total importance
0
20
40
60
80
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread

(a)


weight path
gradient × input
integrated gradients
ablation
88.2
1.6
1.4
4.8
4.0
76.7
2.9
0.6
16.9
3.0
75.8
2.9
0.5
18.0
2.9
38.1
8.6
1.6
47.1
4.5
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
% of total importance
0
30
60
90

(b)